Validation of a Selective Ensemble-Based Classification Scheme for Myoelectric Control Using a Three-Dimensional Fitts' Law Test

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***Abstract*—When controlling a powered upper limb prosthesis it is important not only to know how to move the device, but also when not to move. A novel approach to pattern recognition control, using a selective multiclass one-versus-one classification scheme has been shown to be capable of rejecting unintended motions. This method was shown to outperform other popular classification schemes when presented with muscle contractions that did not correspond to desired actions. In this work, a 3-D Fitts' Law test is proposed as a suitable alternative to using virtual limb environments for evaluating real-time myoelectric control performance. The test is used to compare the selective approach to a state-of-the-art linear discriminant analysis classification based scheme. The framework is shown to obey Fitts' Law for both control schemes, producing linear regression fittings with high coefficients of determination ($R^2 > 0.936$). Additional performance metrics focused on quality of control are discussed and incorporated in the evaluation. Using this framework the selective classification based scheme is shown to produce significantly higher efficiency and completion rates, and significantly lower overshoot and stopping distances, with no significant difference in throughput.**

***Index Terms*—Amputee, electromyogram (EMG), myoelectric, myoelectric signal, pattern recognition, prostheses.**

I. Introduction

PATTERN recognition based myoelectric control has been widely reported in the literature to have great potential for improving the quality and intuitiveness of prosthetic control. Particularly, it has been shown to provide the ability to control many DOFs sequentially without requiring state or mode-switching [1]–[4]. This significant body of work has to date, however, failed to result in any commercially available pattern recognition based myoelectric control schemes. Recently, research has begun to focus on the clinical viability of the pattern recognition based approach and has highlighted the challenges impeding its deployment in clinical settings [5]. These impediments have been speculatively linked to many factors including socket fitting issues [6], the repeatability of signal patterns, and changes in the electromyogram (EMG) patterns over time [7], [8].

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Of particular interest in this work is the repeatability of signal patterns. Traditional pattern recognition studies often constrain the data collection scenario, providing ideal conditions for reporting classification accuracy, but can yield unrealistically repeatable contractions. This may not be representative of clinical usage scenarios which require users to perform task oriented activities under a variety of conditions (loading, position, temperatures, etc.). The resultant EMG is inevitably less constrained and can result in contraction patterns that are unlike those observed during classifier training.

While many classification schemes have been proposed in the literature [9]–[11], the state-of-the-art is generally accepted to be some variant of the system proposed by Hudgins *et al.* [12]. They introduced a set of time-domain (TD) features that have since been widely adopted (due, in part, to their combination of performance and minimal computational burden). For similar reasons, these features are often paired with a simple linear discriminant analysis (LDA) classifier [13]. This TD-LDA combination has been reported frequently in recent studies [5], [11], [14], [15].

The LDA classifier is derived from Bayes' principles and therefore produces a result corresponding to the class with maximum likelihood. While this has been shown to be effective in a constrained system, it also means that anomalous *active* contractions (such as those generated during motion transitions or changes in limb position during unconstrained tasks [14]) often yield erroneous *active* control outputs. These result in unintended activation of the prosthesis which, at a minimum, requires the user to perform compensatory motions and can lead to collisions, dropped objects and increased frustration.

In [16], Scheme *et al.* presented a novel classification scheme based on an ensemble of binary one-versus-one (1vs1) classifiers. Because of its hierarchical nature, this control system can selectively reject active outputs when there is disagreement between the binary classifiers. It was postulated that myoelectric control usability could be improved by forcing the output to no motion during periods of incongruity. The rejection performance was supported by comparing several common classification schemes under nonideal conditions, using prerecorded data. The inclusion of *unknown* contractions was simulated by iteratively excluding one motion class from training while including it during the testing phase. This approach highlighted an important aspect of prosthetic controls research, but still suffered from two weaknesses: 1) the *unknown* motions were not wholly representative of the aberrant motions seen in true usage scenarios (they were actually known, classifiable contractions that were just excluded from training) and 2) the approach still used of-

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fline analysis and did not incorporate the user performing a task, with real time feedback of task performance.

Inclusion of the user as part of the control system via real-time feedback is paramount and likely one of the largest sources of disparity between offline classification studies and clinically observed usability. In 2005, Lock *et al.* [17] asked subjects to perform a virtual clothespin task and found only a weak correlation between classification accuracy and usability. In 2007, Hargrove *et al.* [18] showed improvement in a similar task by including transient contractions in the training data that *decreased* offline classification accuracy. It should be noted, however, that in both of these studies user confusion related to perception of the virtual environment was listed as a potential source of error.

In 2009, Kuiken *et al.* [19] introduced a *motion test* for evaluating classifier performance with user feedback. The test required users to drive a virtual limb through various prompted DOFs. The test, however, ignored incorrect classifications and did not consider proportional control, which is commonly paired with classification schemes in myoelectric control schemes. Simon *et al.* [20] later introduced a *target achievement control* (TAC) test which incorporated these omissions. The TAC test required users to move a virtual limb into a target posture and remain there for a set period of time. Performance was assessed using four main metrics; *classification accuracy*, *completion rate, completion time*, and *path efficiency*.

Upon closer inspection of the TAC test, it can be seen that the task is analogous to a subset of the better known Fitts' Law test. In his pivotal 1954 paper [21], Fitts first quantified human motor performance using principles derived from Shannon's work in communication theory [22]. He demonstrated that any human motor task conveys a finite amount of information that is limited only by the capabilities of the control system and exhibits a tradeoff between speed and accuracy. Fitts' Law testing has been used extensively in the validation of human–computer interfaces (HCI), including various devices such as mice, joysticks, and human motion [23], and forms the basis of an international standard (ISO9341-9).

Fitts' Law has more recently been used to evaluate EMG based HCI control schemes. In 2008, Park *et al.* validated the suitability of EMG as a control source with Fitts' Law [24]. In 2009, Choi *et al.* [25] compared the relative performance of a foreram EMG-based HCI with a standard computer mouse. In 2008, Williams and Kirsch [26] used a Fitts' Law test to evaluate an HCI using EMG signals from neck muscles for individuals with high tretraplegia. Of particular interest in their paper, they presented performance metrics in addition to throughput (the standard summary performance measure obtained from a Fitts' Law test) that closely compare to those used by Simon *et al.* with their TAC test.

Herein, the previously reported 1vs1 selective classifier is compared to a state-of-the-art LDA classification scheme using a Fitts' Law tracking test and evaluated using the metrics proposed by Williams and Kirsch.

II. Methodology

Ten healthy, normally-limbed subjects ranging in age from 19–52 performed a modified 3-D Fitts' Law test. All experiments were approved by the University of New Brunswick's Research Ethics Board.

Subjects were fitted with a cuff made of thermo formable gel that was embedded with six equally spaced pairs of stainless steel dome electrodes. The cuff was placed around the dominant forearm (the right side for all participants), at approximately one third of the length of the forearm at the area of largest muscle bulk. An additional stainless steel dome was included in the cuff as a reference electrode.

The six channels of EMG were differentially amplified using remote ac electrode-amplifiers (BE328 by Liberating Technologies, Inc.) and low pass filtered at 450 Hz with a fifth-order Butterworth filter. Data were sampled with a sampling frequency of 1000 Hz using a 16-bit analog-to-digital converter.

A. Control

Subjects were prompted to elicit contractions corresponding to the following seven classes of motion; *wrist flexion/extension, wrist pronation/supination, hand open/close*, and *no motion*. Four repetitions of 2 s were collected for each motion, during which the subjects dynamically (and subjectively) *ramped* from a low level contraction to a moderately hard level. Subjects were instructed to stay within a comfortable and sustainable range. Including dynamic data during training has been reported to improve classifier robustness [27].

Data were digitally high-pass filtered using a third-order Butterworth filter with a cutoff of 20 Hz in order to remove any motion artifact. Prior to training, data were segmented for feature extraction using 160 ms windows with an overlap of 16 ms. Both the LDA and 1vs1 classifiers were trained using all data collected during training.

Fig. 1 illustrates the real-time data flow of the control scheme used with both classifiers. During testing, as in training, decisions were made every 16 ms using 160 ms windows. Note that a proportional control signal was derived from the same window using the following equation:

$$PC_j = \frac{G_j}{N_{CH}} \sum_{i=1}^{N_{CH}} \sum_{n=1}^{N_S} |x_i[n]| \tag{1}$$

where PC_j is the proportional output for class j, N_S is the number of samples per window, N_{CH} is the number of channels, x_i is the data from channel i, and G_j is a class-specific gain calculated during training. The proportional control output was used to control the velocity or size of the cursor in a given direction, determined by the classifier output.

B. Fitts' Law Task

Although Fitts' Law analysis is most often applied to 2-D tasks, the same principles hold for increased dimensionality [28], [29]. For this study, a pseudo 3-D task was created to allow testing of three degree-of-freedom (DOFs) control schemes. The wrist flexion/extension and wrist rotation DOFs were used to control horizontal and vertical movement, respectively, in the Cartesian plain. The hand open/close DOF controlled the diameter of a circular cursor, as shown in Fig. 2.

The mapping from the prosthetic DOFs used in the TAC test to those used in the Fitts' Law task was found to be intuitive. For the wrist motions, it can be thought of as a simple transformation from polar (rotation about the wrist joint) to Cartesian space.

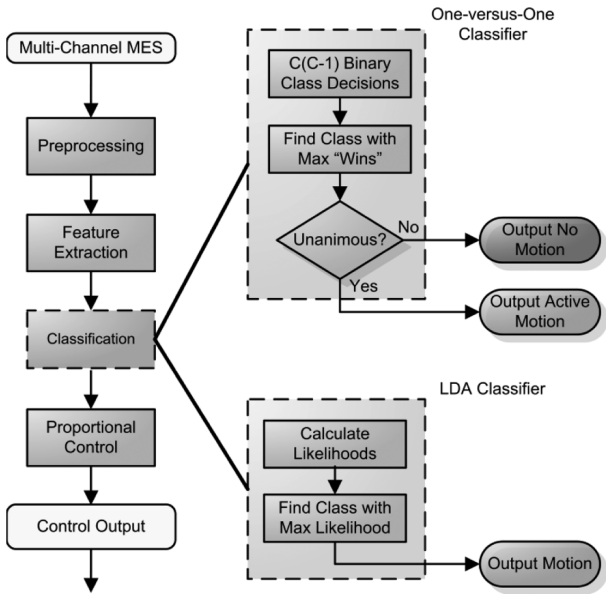

Fig. 1. Block diagram illustrating the data flow of the control schemes used.

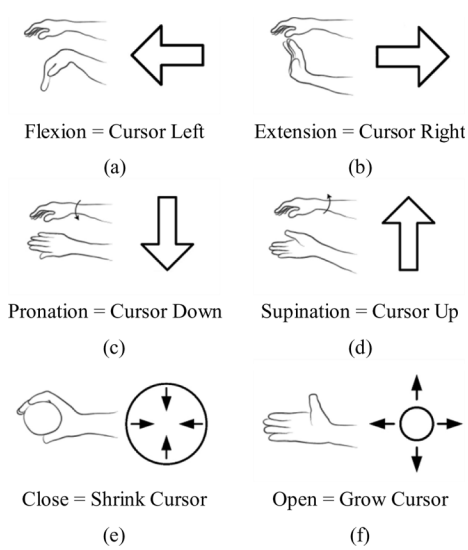

Fig. 2. Mapping from limb motions to controlled DOFs in the Fitts' Law test.

The control of the cursor size is analogous to controlling the aperture of a terminal device.

The Fitts' Law test combines the results from repeated trials, over varying target distances and widths, into a single *throughput* (TP) statistic measured in bits per second. TP is defined as

$$TP = \frac{1}{N}\sum_{i=1}^{N}\frac{ID_i}{MT_i} \tag{2}$$

where i is a particular movement condition, N is the total number of conditions, MT is the time (in seconds) taken to acquire the target, and ID is the index of difficulty (in bits) which relates the target distance D and width W through

$$ID = \log_2\left(\frac{D}{W}+1\right) \tag{3}$$

TABLE I
COMBINATIONS OF DISTANCES (D) AND WIDTHS (W) WITH RESULTING INDICES OF DIFFICULTY (ID)

D	W	ID
50	10	2.59
50	15	2.12
50	25	1.59
100	10	3.46
100	15	2.94
100	25	2.32

where D and W are normalized units of distance.

In their TAC test [20], Simon *et al.* used a movement distance of 75° and a target width of ±5°. This equates to a single ID of 3.09. In practice, Fitts' Law is best served when combining a variety of IDs [30] so, in this study, the distances and widths shown in Table I were used.

In [30], Soukoreff *et al.* suggested an ID range of 2–8 bits, however, that was likely determined for higher resolution, position controlled devices such as mice and trackballs. Pilot work using velocity controlled myoelectric control schemes (where the velocity of the cursor is dictated by the amplitude of the proportional control output) indicated that IDs in excess of 4 resulted in significantly higher trial failure rates and necessitated a departure from the intended ballistic nature of Fitts' Law.

The top speed used during the test (that elicited by a maximal proportional control output) was empirically determined during pilot work. The Fitts' Law task was continually performed while increasing the top speed setting until the observed *throughput* reached a maximum. This was repeated for both classification schemes, and both converged to a relative speed of 1.3 distance units per decision. This was implemented by scaling the value corresponding to a maximum proportional control output from the controller to 1.3 distance units in the graphical interface.

C. Testing Protocol

Users were instructed to complete each *test* as quickly and as naturally as possible. A *test* required the user to move the cursor from a neutral position to a randomly ordered target location (or target size) of distance D and size W. The test was considered complete when the user successfully placed and kept their cursor within the target range for a full second (the *dwell time*). If unsuccessful after 15 s, the test timed out and was considered incomplete. The user cursor was reset to the neutral position between each test.

Tests were completed for all combinations of distance, width, and DOF, resulting in 36 tests per classifier, per trial. After a full trial of both control schemes was completed, the entire process (data collection, classifier training, and Fitts' Law testing) was repeated. This was done a total of 4 times per subject, resulting in a total of 40 separate trials of 36 tests for each classifier.

D. Performance Metrics

While the *throughput* output of the Fitts' Law test is a convenient summary of performance, it does not describe all aspects of the control required to accomplish the task. Simon *et al.* [15] (with their TAC test) and Williams and Kirsch [26] (with their Fitts' Law test) proposed additional metrics that more qualitatively described the nature of the control. This is of particular

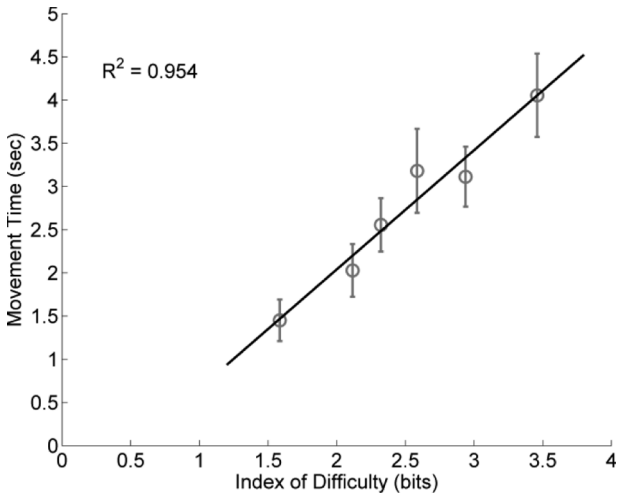

Fig. 3. Relationship between movement time (MT) and index of difficulty (ID) for the LDA-based control scheme.

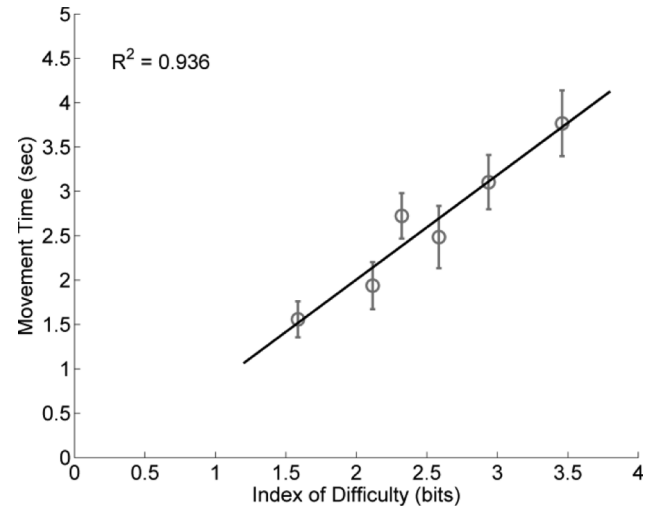

Fig. 4. Relationship between movement time (MT) and index of difficulty (ID) for the 1vs1-based control scheme.

importance when addressing myoelectric control schemes for prosthetic control. The majority of clinically used tests of prosthetic function, such as the Assessment for Capacity of Myoelectric Control (ACMC) [31], rely heavily on qualitative descriptors of the naturalness, spontaneity and compensatory motions during use. In an effort to generate more clinically relevant research, it is important to incorporate similar principles. As such, the performance metrics shown in Table II were included in the analysis.

Statistical analyses of these performance metrics were conducted using a multi-way ANOVA with a post-hoc multiple comparison (*Tukey–Kramer*) test.

III. Results

Figs. 3 and 4 show regression plots of movement time versus index of difficulty for the LDA-based and 1vs1-based control schemes, respectively. These figures show a strong linear relationship between MT and ID for both control schemes, producing high coefficients of determination ($R^2_{\mathrm{LDA}} = 0.954, R^2_{\mathrm{1vs1}} = 0.936$). This strongly supports the suitability of Fitts' Law testing for velocity-based myoelectric controls research.

Table III summarizes the performance of the control schemes for the various performance metrics, averaged across all users and trials. The LDA-based control scheme produced a slightly higher average *throughput* value ($\mathrm{TP}_{\mathrm{LDA}} = 1.10 \pm 0.03$, $\mathrm{TP}_{\mathrm{1vs1}} = 1.07 \pm 0.02$), although no statistical difference was observed ($p = 0.242$). The 1vs1-based classifier produced significantly better results for all other performance metrics. It yielded more than 10% greater *efficiency* than the LDA-based control scheme ($\mathrm{EFF}_{\mathrm{LDA}} = 77.09 \pm 0.89$, $\mathrm{EFF}_{\mathrm{1vs1}} = 87.4 \pm 0.52, p < 0.001$). The 1vs1-based scheme's *overshoot* per test was less than half of that of the LDA-bases scheme ($\mathrm{OS}_{\mathrm{LDA}} = 0.56 \pm 0.04$, $\mathrm{OS}_{\mathrm{1vs1}} = 0.21 \pm 0.01$, $p < 0.001$). The *stopping distance* was nearly 25% less ($\mathrm{SD}_{\mathrm{LDA}} = 438.65 \pm 16.03$, $\mathrm{SD}_{\mathrm{1vs1}} = 337.46 \pm 10.72$, $p < 0.001$), and the completion rate was significantly higher ($\mathrm{CR}_{\mathrm{LDA}} = 93.44 \pm 0.78$, $\mathrm{CR}_{\mathrm{1vs1}} = 96.1 \pm 0.57$, $p = 0.002$).

TABLE II
Performance Metrics Used

Metric	*Description*
Throughput	Describes usability through the tradeoff of speed and accuracy
Efficiency	Describes the systems quality of control; a ratio of the shortest path to the target to the actual path travelled
Overshoot	Describes the ability to stop on a target; the average number of times the target was acquired and then lost per test
Stopping Distance	Describes the ability to hold no motion; the total distance travelled (path length) during the 1 second dwell time
Completion Rate	Describes overall success; the percentage of tests completed within the allowed time

TABLE III
Results of the Three Dimensional Fitts' Law Test

	LDA	*1vs1*	*P-value*
Throughput	1.10 ± 0.03	1.07 ± 0.02	p=0.242
Efficiency (%)	77.09 ± 0.89	87.4 ± 0.52	p<0.001
Overshoot	0.56 ± 0.04	0.21 ± 0.01	p<0.001
Stopping Distance	438.65 ± 16.03	337.46 ± 10.72	p<0.001
Completion Rate (%)	93.44 ± 0.78	96.1 ± 0.57	p=0.002

Fig. 5 shows an example of the DOF position trajectories during a test with 100% *efficiency* and no *overshoot*. Note that once the wrist rotation DOF was engaged it remained active until the target was acquired (as evidenced by the continuously

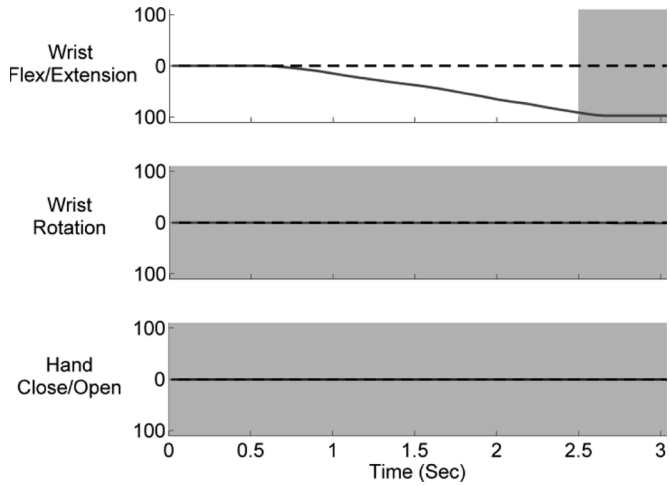

Fig. 5. Position trajectories from of a completed wrist flexion test with near 100% efficiency. The vertical axes represent the distance travelled from neutral in a particular direction. Shaded areas indicate that the target was acquired for that DOF.

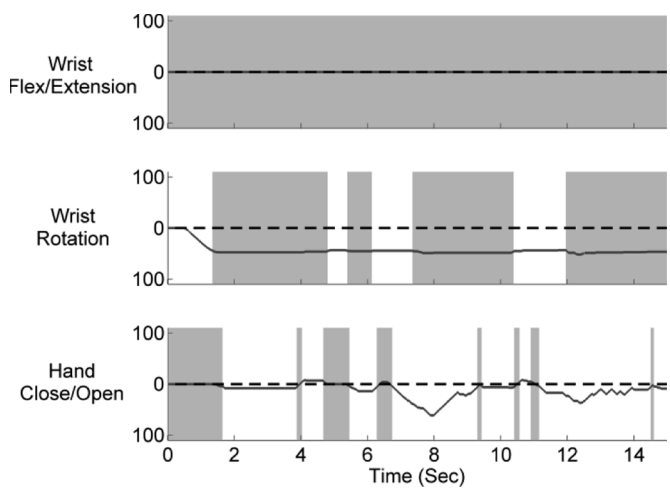

Fig. 6. Position trajectories of a test of the LDA-based scheme requiring wrist rotation when a timeout occurred. The vertical axes represent the distance travelled from neutral in a particular direction. Shaded areas indicate that the target was acquired for that DOF.

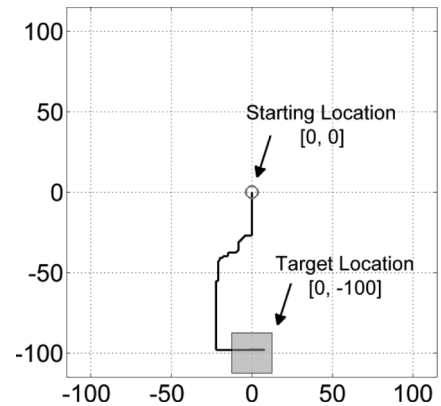

Fig. 7. 2-D example of a completed LDA test that required correction for unintentional flexion.

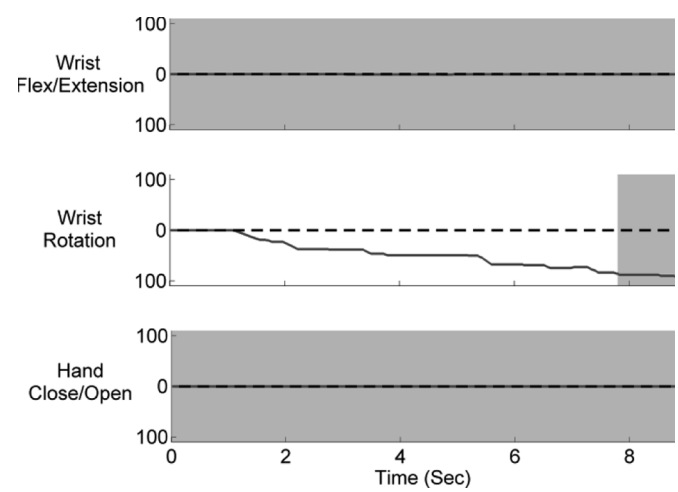

Fig. 8. Position trajectories of a test of the 1vs1-based scheme requiring wrist pronation that exhibited excessive rejection. The vertical axes represent the distance travelled from neutral in a particular direction. Shaded areas indicate that the target was acquired for that DOF.

positive slope). The wrist flexion/extension and hand close/open DOFs were inactive during the entire test.

Fig. 6 shows an example of the DOF position trajectories during an unsuccessful test of the LDA-based scheme. The oscillatory nature of the hand close/open DOF was due to unintentional and uncontrollable activation of the hand close class during intended no motion, and made remaining on target challenging.

Fig. 7 depicts a 2-DOF example of a successful test of the LDA-based scheme (the third DOF has been omitted for clarify). Note that a 100% *efficient* test would have elicited a purely vertical trace from the center to the target. During this test, the control scheme produced several erroneous *left* commands (wrist flexion) which the user corrected after completing the vertical requirements of the task.

Fig. 8 shows an example of position trajectories during a test of the 1vs1-based scheme with excessive rejection. In this case, the subject was unable to sustain the wrist pronation contraction without causing rejection (shown by the sections of zero change in the wrist rotation position). The test was completed successfully because the user resorted to a pulsatile style of control. This is an example of how, even in cases of excessive rejection, the overall appearance of the 1vs1-based control is smoother.

Figs. 9 and 10 show 3-D renderings of the trajectories taken by a user while using the LDA-based and the 1vs1-based schemes, respectively. These results represent all tests with a D of 100 and W of 10, 15, or 25. Note that a straight line from location $[0, 0, 0]$ to one of the target boxes represents a 100% efficient test and so any deviation from these lines indicates that an error occurred which caused the user to deviate from the optimal path. Note that any trajectory line that intersects two planes of a target denotes that an overshoot occurred.

IV. Discussion

The Fitts' Law framework has been used extensively in other fields for many years and is a well-defined and well-documented metric for the evaluation of motor control schemes. The task requirements, which involve acquisition of a target location through control of a cursor, are a close analog to previously reported myoelectric control studies. While the

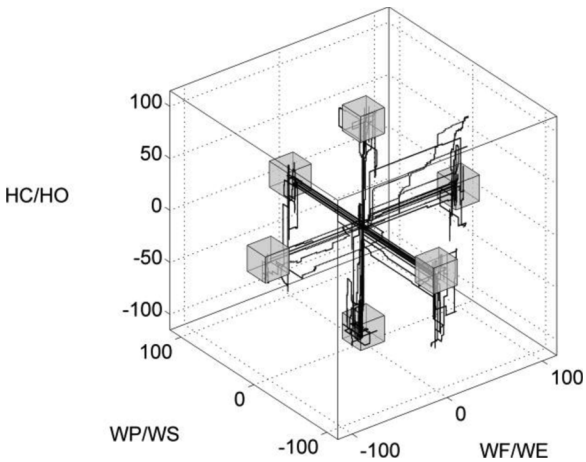

Fig. 9. Path traces from a representative user including all tests with $D = 100$ and $W = [10, 15, 25]$ for the LDA-based scheme.

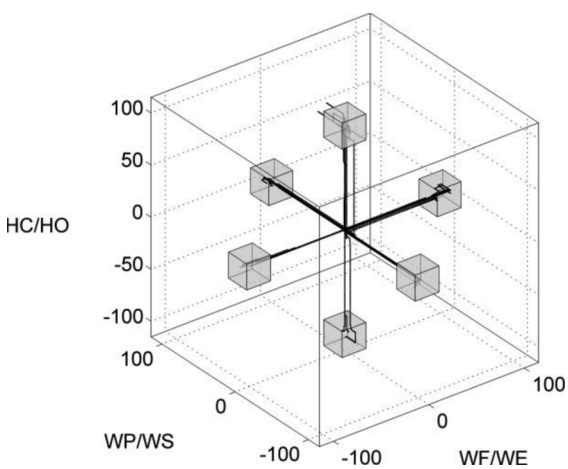

Fig. 10. Path traces from a representative user including all tests with $D = 100$ and $W = [10, 15, 25]$ for the 1vs1-based scheme.

use of virtual environments for simulation of the control of prosthetic devices is appealing, their design is cumbersome and many suffer from limitations in performance, visualization and immersion. The design requirements for pattern recognition based myoelectric control schemes remain similar whether they are used for prosthetic control or HCIs. As such, it makes sense to exploit standardized testing methods adopted by other fields of research. The results of Figs. 3 and 4 support the use of Fitts' Law for velocity based myoelectric control schemes due to the high R^2 correlation with its log-relationship between movement time and the distance/width ratio.

When considering prosthetic control applications, the naturalness of control also plays an important role. Clinical evaluation of prosthetics is rarely based wholly on completion time and often describes aspects such as the smoothness and efficiency of the control. The inclusion of additional performance metrics such as those proposed by Williams and Kirsch [26], and Simon *et al.* [32] are therefore of particular importance.

In this study, two different methods of pattern recognition based myoelectric control were compared. The LDA-based scheme, commonly used in prosthetics research, produced similar *throughput* to that of the rejection-capable 1vs1-based scheme. The 1vs1-based scheme, however, significantly outperformed the LDA-based scheme in all other performance metrics. From this, two things can be inferred. The first is that the rejection component of the 1vs1 classifier is yielding considerable improvement in the quality of control, given the vast improvement in metrics such as *efficiency* and *completion rate*. Conversely, the second is that the 1vs1-based scheme is too frequently rejecting active motion. The *efficiency* metric is determined using path efficiency (not time efficiency, which is incorporated in the *throughput* metric), and consequently does not incorporate penalties for no motion decisions. In theory, rejecting only those decisions that would be misclassified by the LDA-based scheme should yield improvement in both *efficiency* and *throughput* (because of the elimination of the need for resultant compensatory motions). The approximately 10% improvement in *efficiency* coupled with no significant change in *throughput* therefore implies that the 1vs1-based scheme rejected not only erroneous decisions, but also some correct decisions. Future work in optimizing this false activation versus false rejection tradeoff could produce an improved system with higher *throughput*.

Some subjects reported frustration with the LDA-based scheme, describing a "bounce-back" behavior. This commonly occurred when, while attempting to stop an active contraction, the opposing motion would be activated. This was the primary cause of incomplete trials due to oscillation about a target caused by alternating errors and attempted compensation. In some subjects, this effect was somewhat reduced by coaching them to focus on stopping the contraction instead of focusing on "returning to neutral."

The most commonly reported observation of the 1vs1-based scheme was that it felt more controlled, but slower. The proportional control algorithm for both schemes was identical, so the ostensible speed differences were likely caused by intermittent rejection. Because decisions were made very quickly (every 16 ms), sporadic rejections were too short to be uniquely identified in real-time and instead caused a perceived reduction in velocity. This type of rejection behavior is more acceptable than that exhibited in Fig. 8 and, if repeatable, may be alleviated by increasing the maximum speed for this type of control or including some form of smoothing (majority vote) to eliminate spurious rejections.

Finally, the proposed framework evaluates control schemes based on their ability to properly predict direction (classification, in pattern recognition based approaches) but also on their ability to control speed (proportional control). In ongoing research, the authors are looking at further optimizing the use of rejection for pattern recognition based control as well as new methods for improving proportional control. It was shown in [16] that the relative performance of various classification schemes was consistent between able bodied and amputee populations. This validated using able bodied subjects to compare offline classification schemes, but this result has yet to be confirmed for real-time usability studies with amputees. While the results of this study have shown significant improvement in control for able bodied subjects, the performance of the selective rejection approach with amputee subjects in a clinical setting should be verified. As a result, future work should

include testing with amputee subjects while wearing custom fitted sockets.

V. Conclusion

A Fitts' Law-based usability test was proposed as an alternative to virtual limb environments for the evaluation of myoelectric prosthetic control schemes. The resultant *throughput* summary statistic was combined with more illustrative performance metrics (*efficiency*, *overshoot, stopping distance*) and *completion rate* to provide a more complete description of the control performance. High R^2 values obtained from regression plots of the Fitts' Law task confirmed that it is viable as a usability testing tool for myoelectric control research. The framework was used to compare the performance of a previously reported selective 1vs1 classification based control scheme to a state-of-the-art LDA-based scheme. The 1vs1-based scheme was previously shown to outperform several commonly used classifiers under non-ideal offline testing conditions. In this study, the 1vs1-based control scheme produced significantly higher *efficiency* $(p < 0.001)$ and *completion rates* $(p = 0.002)$, and significantly lower *overshoot* $(p < 0.001)$ and *stopping distances* $(p < 0.001)$ when compared to the LDA-based scheme. No significant difference was observed in *throughput* values $(p = 0.242)$.

The significant improvement in all performance areas, with the exception of throughput, indicates that the rejection-capable 1vs1-based scheme may represent a smoother, more controlled option for myoelectric-controlled prostheses.

References

[1] P. Parker, K. Englehart, and B. Hudgins, "Myoelectric signal processing for control of powered prosthesis," *J. Electromyogr. Kinesiol.*, vol. 16, no. 6, pp. 541–548, Dec. 2006.

[2] D. Graupe, J. Salahi, and K. Kohn, "Multifunctional prosthesis and orthosis control via microcomputer identification of temporal pattern differences in single-site myoelectric signals," *J. Biomed. Eng.*, vol. 4, pp. 17–22, Jan. 1982.

[3] L. Hargrove, K. Englehart, and B. Hudgins, "A comparison of surface and intramuscular myoeletric signal classification," *IEEE Trans. Biomed. Eng.*, vol. 54, no. 5, pp. 847–853, May 2007.

[4] M. Oskoei, "Myoelectric control systems—A survey," *Biomed. Signal Process. Control*, vol. 2, pp. 275–294, 2007.

[5] E. Scheme and K. Englehart, "EMG pattern recognition for the control of powered upper-limb prostheses: State of the art and challenges for clinical use," *J. Rehabil. Res. Develop.*, vol. 48, no. 6, pp. 643–660, 2011.

[6] A. Boschmann, P. Kaufmann, M. Platzner, and M. Winkler, "Towards multi-movement hand prostheses: Combining adaptive classification with high precision sockets," in *Proc. 2nd Eur. Conf. Technically Assisted Rehabilitation (TAR'09)*, Berlin, Germany, 2009.

[7] J. Sensinger, B. Lock, and T. Kuiken, "Adaptive pattern recognition of myoelectric signals: Exploration of conceptual framework and practical algorithms," *IEEE Trans Neural Syst. Rehabil. Eng.*, vol. 17, no. 3, pp. 270–278, Jun. 2009.

[8] K. Biron and K. Englehart, "EMG pattern recognition adaptation," presented at the 48th Congress Int. Soc. Electrophysiol. Kinesiol., Aalborg, Denmark, Jun. 2010.

[9] Y. H. Huang, K. Englehart, B. Hudgins, and A. Chan, "A Gaussian mixture model based classification scheme for myoelectric control of powered upper limb prostheses," *IEEE Trans. Biomed. Eng.*, vol. 52, no. 11, pp. 1801–1811, Nov. 2005.

[10] M.-F. Lucas, A. Gaufriau, S. Pascual, C. Doncarli, and D. Farina, "Multi-channel surface EMG classification using support vector machines and signal-based wavelet optimization," *Biomed. Signal Process. Control*, vol. 3, pp. 169–174, 2008.

[11] L. Hargrove, E. Scheme, K. Englehart, and B. Hudgins, "Multiple binary classifications via linear discriminant analysis for improved controllability of a powered prosthesis," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 18, no. 1, pp. 49–57, Jan. 2010.

[12] B. Hudgins, P. Parker, and R. N. Scott, "A new strategy for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 40, no. 1, pp. 82–94, Jan. 1993.

[13] K. Englehart and B. Hudgins, "A robust, real-time control scheme for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 7, pp. 848–854, Jul. 2003.

[14] A. Fougner, E. Scheme, A. Chan, K. Englehart, and Ø. Stavdahl, "Resolving the limb position effect in myoelectric pattern recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 19, no. 6, pp. 644–651, Dec. 2011.

[15] A. Simon, B. Lock, K. Stubblefield, and L. Hargrove, "Prosthesis-Guided training for practical use of pattern recognition control of prostheses," presented at the Myoelectric Control Symp., Fredericton, NB, Canada, 2011.

[16] E. Scheme, K. Englehart, and B. Hudgins, "Selective classification for improved robustness of myoelectric control under nonideal conditions," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 6, pp. 1698–1705, Jun. 2011.

[17] B. Lock, K. Englehart, and B. Hudgins, "Real-Time myoelectric control in a virtual environment to relate usability vs. Accuracy," presented at the Myoelectric Controls Symp., Fredericton, NB, Canada, 2005.

[18] L. Hargrove, Y. Losier, B. Lock, K. Englehart, and B. Hudgins, "A real-time pattern recognition based myoelectric control usability study imlemented in a virtual environment," in *Proc. IEEE Int. Eng. Med. Biol. Soc. Conf.*, Aug. 2007, pp. 4842–4845.

[19] T. Kuiken, L. Guanglin, B. Lock, R. Lipschutz, L. Miller, K. Stubblefield, and K. Englehart, "Targeted muscle reinnervation for real-time myoelectric control of multifunction artificial arms," *J. Am. Med. Assoc.*, vol. 301, no. 6, pp. 619–628, 2009.

[20] A. Simon, L. Hargrove, B. Lock, and T. Kuiken, "Target achievement control test: Evaluating real-time myoelectric pattern-recognition control of multifunctional upper-limb prostheses," *J. Rehabil. Res. Develop.*, vol. 48, no. 6, pp. 619–628, 2011.

[21] P. Fitts, "The information capacity of the human motor system in controlling the amplitude of movement," *J. Exp. Psychol.*, vol. 47, pp. 381–391, 1954.

[22] C. Shannon, "Mathematical theory of communication," *Bell Syst. Tech. J.*, vol. 27, pp. 623–656, 1948.

[23] R. Andres and K. J. Hartung, "Prediction of head movement time using Fitts' Law," *Human Factors*, vol. 31, pp. 703–713, 1989.

[24] J. Park, W. Bei, H. Kim, and S. Park, "EMG—Force correlation considering Fitts' Law," in *IEEE Int. Conf. Multisensor Fusion Integration Intell. Syst.*, Aug. 2008, pp. 644–649.

[25] C. Choi, S. Micera, J. Carpaneto, and J. Kim, "Development and quantitative performance evaluation of a noninvasive EMG computer interface," *IEEE Trans. Biomed. Eng.*, vol. 56, no. 1, pp. 188–191, Jan. 2009.

[26] M. Williams and R. Kirsch, "Evaluation of head orientation and neck muscle EMG signals as command inputs to a human-computer interface for individuals with high tetraplegia," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 16, no. 5, pp. 485–496, Oct. 2008.

[27] E. Scheme and K. Englehart, "Examining the effect of proportional control on classification in pattern recognition based myoelectric control," *IEEE Trans. Neural Syst. Rehabil. Eng.*, submitted for publication.

[28] A. Murata and H. Iwase, "Extending Fitts' Law to a three-dimensional pointing task," *Hum. Movement Sci.*, vol. 20, no. 6, pp. 791–805, 2001.

[29] J. Vaughan, D. Barany, A. Sali, S. Jax, and D. Rosenbaum, "Extending Fitts' Law to three-dimensional obstacle-avoidance movements: Support for the posture-based motion planning model," *Exp. Brain Res.*, vol. 207, no. 1–2, pp. 133–138, Nov. 2010.

[30] R. Soukoreff and I. MacKenzie, "Towards a standard for pointing device evaluation, perspectives on 27 years of Fitts' Law research in HCI," *Int. J. Human-Comput. Studies*, vol. 61, no. 6, pp. 751–789, Dec. 2004.

[31] L. Miller and S. Swanson, "Summary and recommendations of the academy's state of the science conference on upper limb prosthetic outcome measures," *J. Prosthetics Orthotics*, vol. 21, no. 9, pp. 83–89, 2009.

[32] A. Simon, K. Stern, and L. Hargrove, "A comparison of proportional control methods for pattern recognition control," in *Proc. IEEE Int. Symp. Eng. Med. Biol. Soc. Conf.*, 2011, pp. 3354–3357.

[33] A. H. Bottomley, "Working model of a myoelectric control system," in *Proc. Int. Symp. Appl. Autom. Control Prosthetics Design*, 1962, pp. 37–37.
[34] P. Herbets, "Myoelectric signals in control of prostheses," *Acta. Orthop. Scand.*, 1969, Suppl. 124.
[35] D. S. Dorcas and R. N. Scott, "A three-state myoelectric control," *Med. Biol. Eng.*, vol. 4, pp. 367–372, 1966.
[36] R. R. Finley and R. W. Wirta, "Myocoder studies of multiple myocoder response," *Arch. Phys. Med. Rehab.*, vol. 48, pp. 598–598, 1967.
[37] N. Hogan and R. Mann, "Myoelectric signal processing: Optimal estimation applied to electromyography—Part 1: Derivation of the optimal myoprocessor," *IEEE Trans. Biomed. Eng.*, vol. 27, no. 7, pp. 382–395, Jul. 1980.
[38] H. Evans, Z. Pan, P. Parker, and R. Scott, "Signal processing for proportional myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 31, no. 2, pp. 207–211, Feb. 1984.
[39] E. Scheme, A. Fougner, Ø. Stavdahl, A. Chan, and K. Englehart, "Examining the adverse effects of limb position on pattern," in *Proc. IEEE Eng. Med. Biol. Soc. Conf.*, 2010, pp. 6337–6340.
[40] L. Hargrove, K. Englehart, and B. Hudgins, "The effect of electrode displacements on pattern recognition based myoelectric control," in *Proc. IEEE Eng. Med. Biol. Soc. Conf.*, 2006, pp. 2203–2206.
[41] L. Hargrove, K. Englehart, and B. Hudgins, "A training strategy to reduce classification degradation due to electrode displacements in pattern recognition based myoelectric control," *Biomed. Signal Process. Control*, vol. 3, pp. 175–180, 2008.
[42] A. Young, L. Hargrove, and T. Kuiken, "The effects of electrode size and orientation on the sensitivity of myoelectric pattern recognition systems to electrode shift," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 9, pp. 2537–2543, Sep. 2011.
[43] E. Corbett, E. Perreault, and T. Kuiken, "Comparison of electromyography and force as interfaces for prosthetic control," *J. Rehabil. Res. Develop.*, vol. 48, no. 6, pp. 629–642, 2011.

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